

PROBLEM SHEET 4: SOLUTIONS & FEEDBACK

1. (a) The homogeneous system is

$$\frac{du}{dt} = u(1 - u - v), \quad (1a)$$

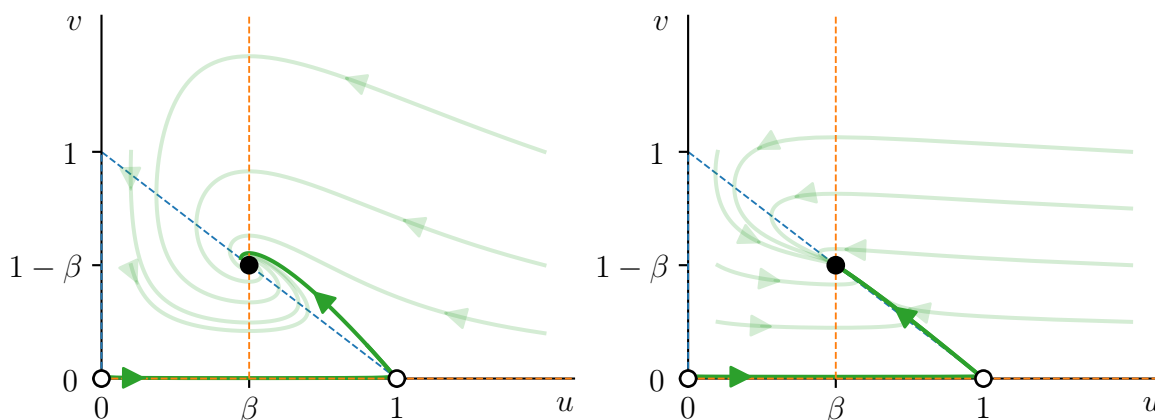
$$\frac{dv}{dt} = \alpha v(u - \beta), \quad (1b)$$

with nullclines

$$\frac{du}{dt} = 0 \implies u = 0, \quad v = 1 - u \quad (2a)$$

$$\frac{dv}{dt} = 0 \implies u = \beta, \quad v = 0. \quad (2b)$$

The phase portrait looks like:



where the case with $(\beta, 1 - \beta)$ as a stable spiral is on the left, and the case with $(\beta, 1 - \beta)$ as a stable node is on the right. Note on the right that the trajectory from $(1, 0)$ to $(\beta, 1 - \beta)$ does not go directly along the nullcline: if you zoom in, you can see it is slightly to the right (which makes sense, given that trajectories cross the nullcline vertically).

 Feedback

Drawing the nullclines was done fine, but the trajectories in general were drawn very badly. We covered this extensively in Problems Class 2: if you are not showing up to problems classes, consider this your warning! Common errors leading to lost marks included:

- i. Not drawing trajectories as continuous lines. This is not like drawing a vector field in AMV or a velocity field in Fluids, where you just draw little arrows to represent the direction/magnitude at some points: trajectories are paths and they have to start somewhere and end either at an equilibrium, or go off to infinity.
- ii. Not having the trajectory curves cross the nullclines either vertically or horizontally. This is *crucial* to getting the right shape and you will lose

marks in an exam if you don't draw these accurately.

- iii. Having trajectory curves change either vertical or horizontal direction away from a nullcline. This is not allowed; that is the point of the nullclines: the locations where they change direction.
- iv. Not drawing a diagram that is big enough to contain all the detail you need. Go big or go home! In some cases, this led people to drawing trajectories which cross, which, you will remember from Problems Class 2, is not allowed.
- v. Not observing there are two possibilities. The question gives you a clear steer on this by highlighting the two: that means two diagrams. The local structure of the linear system (node or spiral) has to dictate how you draw the curves near the equilibrium.

- (b) Since $(0, 0)$ is a saddle point, there is only one trajectory leaving it with $u > 0$ and we can see that it corresponds to the horizontal nullcline, i.e. trajectories leave $(0, 0)$ and go to $(1, 0)$. Therefore, there is no direct trajectory from $(0, 0)$ to $(\beta, 1 - \beta)$.

Feedback

There were quite a lot of misunderstandings about trajectories. A common error was saying that because you had seen the trajectories spiral in that a path exists. But, there are a couple of issues. First a travelling wave, the way we define it in class, has to go from an equilibrium to an equilibrium. If you are at $(0, 0)$ the only way out is along the saddle arc, which is along $v = 0$. Since this is a nullcline, $dv/dt = 0$ always, and so the trajectory will end at $(1, 0)$. It is fine to observe that in the sector between $(0, 0)$ and $(\beta, 1 - \beta)$ the trajectories have to head downwards, so you can't go towards $(\beta, 1 - \beta)$ anyway. But then you have to explain why the trajectories can't leave the u -axis after the line $u = \beta$.

2. (a) If the stable equilibrium is at $x = \infty$, we would expect this region to expand and therefore the wave to travel *leftwards*, i.e. $z = x + ct$ with $c > 0$.

Feedback

The main error was switching x to $-x$. The locations of the particular equilibria are fixed in the question, so you can't swap: it has to be c which changes sign.

- (b) We look for solutions in z where

$$u(x, t) = u(z), \quad v(x, t) = v(z), \quad z = x - ct. \quad (3)$$

Substituting z into ( ), we obtain

$$-c \frac{du}{dz} = u(1 - u - v), \quad (4a)$$

$$-c \frac{dv}{dz} = \frac{d^2v}{dz^2} + \alpha v(u - \beta). \quad (4b)$$

Introducing $w = dv/dz$ and rearranging, we arrive at the three-dimensional

system

$$\frac{du}{dz} = \frac{u}{c}(u + v - 1), \quad (5a)$$

$$\frac{dv}{dz} = w, \quad (5b)$$

$$\frac{dw}{dz} = \alpha v(\beta - u) - cw. \quad (5c)$$

The boundary conditions are

$$(u, v, w) \rightarrow (\beta, 1 - \beta, 0) \text{ as } z \rightarrow -\infty, \quad (6)$$

$$(u, v, w) \rightarrow (1, 0, 0) \text{ as } z \rightarrow \infty. \quad (7)$$

Feedback

Two main errors:

- i. Putting the stable equilibrium at $z \rightarrow \infty$: the movement should be away from that as it slowly takes over.
- ii. Only giving one boundary condition on u . You need to also give a boundary condition on w .

3. (a) The Jacobian of the system (  ) is

$$J = \begin{pmatrix} (2u + v - 1)/c & u/c & 0 \\ 0 & 0 & 1 \\ -\alpha v & \alpha(\beta - u) & -c \end{pmatrix}. \quad (8)$$

(b) The Jacobian, evaluated at $(1, 0, 0)$ gives us

$$J = \begin{pmatrix} 1/c & 1/c & 0 \\ 0 & 0 & 1 \\ 0 & \alpha(\beta - 1) & -c \end{pmatrix}. \quad (9)$$

This matrix leads to the eigenvalue equation

$$(\lambda - 1/c)[\lambda(c + \lambda) - \alpha(\beta - 1)] = 0, \quad (10)$$

and hence the eigenvalues are

$$\lambda = \frac{1}{c}, \quad \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4\alpha(\beta - 1)} \right). \quad (11)$$

Feedback

Some small errors in the matrix calculation; otherwise this was done well.

- (c) The first eigenvalue is positive so we expect an unstable solution. But what type of unstable solution? From the second expression, we see that the solution will be oscillatory (a spiral) if $c^2 < 4\alpha(1 - \beta)$. Since v cannot be less than zero, these oscillatory solutions can be discarded and this gives us a constraint on the wavespeed, just like it did with the FK equation. Namely, the permissible wave speeds satisfy

$$c^2 \geq 4\alpha(1 - \beta). \quad (12)$$

Feedback

Quite a few people missed the key point here. A travelling wave is really just a solution in the form $u(x - ct)$, with the linear coupling of the two independent variables. Its name comes from the wave equation,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad (13)$$

where any $f(x - ct)$ solves the system, whatever the shape of f . So there are no issues with the oscillatory/spiral shape in principle (mathematically). But on *physical* grounds, spirals imply a negative population of v as the spiral is around $(1, 0, 0)$ (so you should say that v is the problem). This was the same argument we used in class for the FK equation.

4. (a) The Jacobian, evaluated at $(\beta, 1 - \beta, 0)$ is

$$\mathbf{J} = \begin{pmatrix} \beta/c & \beta/c & 0 \\ 0 & 0 & 1 \\ \alpha(\beta - 1) & 0 & -c \end{pmatrix}, \quad (14)$$

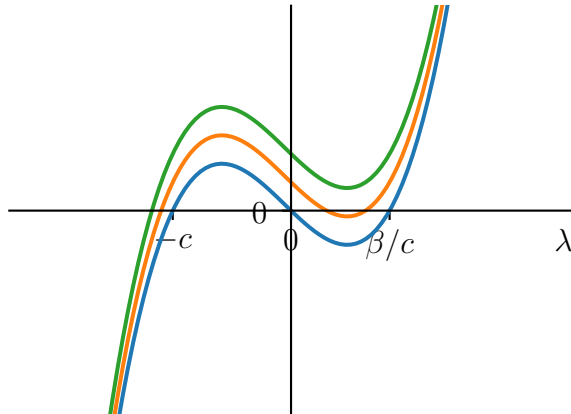
which has eigenvalue equation

$$\lambda^3 + \lambda^2 \left(c - \frac{\beta}{c} \right) - \beta\lambda + \frac{\beta}{c}\alpha(1 - \beta) = 0. \quad (15)$$

Feedback

This was done fine.

- (b) The cubic looks like



for three different values of α . (You do not need to find the points where it crosses the x -axis at this stage.)

Differentiating and setting to zero, we find

$$3\lambda^2 + 2\lambda \left(c - \frac{\beta}{c} \right) - \beta = 0, \quad (16)$$

and solving the quadratic we find

$$\lambda_{\max}, \lambda_{\min} = -\frac{1}{3} \left[c - \frac{\beta}{c} \pm \sqrt{\left(c - \frac{\beta}{c} \right)^2 + 3\beta} \right]. \quad (17)$$

Feedback

This was also done fine. Big up to those who drew snowfolk.

- (c) I am convinced. Looking at eq. (15), changing α is like doing ' $f(x) + a$ ', translating the function vertically. (See picture above.)

Feedback

Many of you seemed convinced. Good for you: back yourselves!

- (d) Equation () with $\alpha = 0$ has solutions $\lambda = 0$ and

$$\lambda^2 + \lambda \left(c - \frac{\beta}{c} \right) - \beta = 0 \quad (18)$$

$$\implies (\lambda + c) \left(\lambda - \frac{\beta}{c} \right) = 0 \implies \lambda = -c, \frac{\beta}{c}. \quad (19)$$

The eigenvalues are therefore $\lambda = 0, -c, \beta/c$.

Feedback

The majority of you used the quadratic formula, which does simplify to these solutions. But you should always have a quick look as to whether you can factorise clearly: in this case all you need is $(a - b)^2 + 4ab = (a + b)^2$. The solutions from the quadratic equation do not simplify obviously and the clarity of the result is important for interpretation.

- (e) Previously, we discarded these solutions on the grounds that they would yield negative values of a population size. In those instances, however, we were dealing with equilibria where one population size was zero. Here, we have $(\beta, 1 - \beta)$, so the solution may safely oscillate without ever becoming negative. Thus, we can permit oscillatory wavefronts. While this is a fundamental difference with the FK equation where oscillations were not possible, like the FK equation the wave speed is given by the minimum possible value, namely, $c^2 = 4\alpha(1 - \beta)$.

 Feedback

Here the issues were similar to the previous question about validity of travelling waves. The issue in the previous case was forbidding negative v ; but that is not an issue here.